SEE MODEL QUESTION 2080

Subject: Science and Technology (Set 2 / Set B)
Complete Solutions Collection (English Medium Only)

Group 'A' - Multiple Choice Questions (10 x 1 = 10)

a) Which is the smallest unit of computer memory?

Answer: (ii) Bit

b) Why are pines kept in sub-division gymnosperm?

Answer: (ii) It has naked seeds.

c) Which of the following groups of organism are closely related on the basis of evolution?

Answer: (iv) Platyhelminthes, Nemathelminthes, Annelida

d) What is the main cause of increasing the level of sea?

Answer: (iv) Increase in the temperature of the earth.

e) Calculate the mass that a person can lift on moon if he can lift a mass of 50 kg on the earth.

Answer: (i) 50 kg
Explanation: Mass is an intrinsic property of matter and remains constant everywhere in the universe regardless of gravitational changes.

f) If large piston A and small piston B of a hydraulic lift have cross-sectional area, pressure, and force as A1, A2; P1, P2; and F1, F2 respectively, then which relation is correct according to Pascal's law?

Answer: (iii) F1 > F2
Explanation: Pressure is transmitted equally (P1 = P2). Since F1/A1 = F2/A2 and area A1 > A2, the output force F1 must be greater than F2.

g) Identify the characteristics of the image of the object XY kept in front of the lens as shown in the given figure.

Answer: (iv) Real, inverted, and diminished (Note: The printed paper has minor textual typos, but this is the standard convex lens feature when an object is placed beyond 2F).

h) Which of the following is proved by Hubble's law?

Answer: (ii) Universe is expanding.

i) Which one is a double displacement reaction from the following?

Answer: (iv) XY + AB -> XB + AY

j) Statement: Detergent is called a soapless soap.
Argument 1: It has cleaning properties like soap but its chemical nature is different.
Argument 2: It does not give lather with hard water as soap.

Answer: (ii) Statement and argument 1 is correct but argument 2 is incorrect.
Explanation: Detergents maintain high performance and lather easily even in hard mineral water, unlike standard soap.

Group 'B' - Very Short Answer Type Questions (9 x 1 = 9)

a) Write an advantage of independent variable.

Answer: It allows the experimental researcher to directly alter specific conditions to observe their exact impacts on a dependent output factor.

b) What is upthrust?

Answer: The total resultant upward force exerted by a fluid on an object completely or partially immersed in it is termed upthrust.

c) Write down one utility of optical fiber.

Answer: It is used in telecommunication systems to transmit digital high-speed signals over massive distances with minimal attenuation via total internal reflection.

d) In summer season, the cemented roads are seemed to be filled with water but they are not actually. Which phenomenon of light is behind this?

Answer: Total Internal Reflection of light (Mirage effect).

e) What happens when carbon dioxide gas is passed through lime water for short time?

Answer: The clear solution turns milky due to the chemical precipitation of insoluble calcium carbonate (CaCO3).

f) What is meant by redshift?

Answer: The displacement of cosmic spectral lines toward longer wavelengths (the red end), indicating that a distant light source is receding due to cosmic expansion.

g) Write the name of main ore of silver.

Answer: Argentite (Ag2S).

h) Why are group VII elements of periodic table called halogens?

Answer: Because they directly interact with reactive metals to produce mineral salts ('halo' means salt, 'gen' means producer).

i) Write the structural formula of glycol.

Answer: Ethane-1,2-diol (Glycol) structural framework layout:

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    H    H
    |    |
H - C - C - H
    |    |
   OH   OH
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Group 'C' - Short Answer Type Questions (14 x 2 = 28)

3. Use fundamental units to show that equation P = IV is homogeneous, where I is electric current, V is voltage and P is power.

Answer:
- **LHS (Power, P):** Unit is Watt = Joule / second = (kg*m^2*s^-2) / s = **kg*m^2*s^-3**
- **RHS (I * V):** Unit of Current (I) = Ampere (A).
Unit of Voltage (V) = Work / Charge = Joule / (Current * Time) = (kg*m^2*s^-2) / (A*s) = kg*m^2*s^-3*A^-1.
Hence, Unit of IV = A * (kg*m^2*s^-3*A^-1) = **kg*m^2*s^-3**
Since LHS matches RHS exactly, the equation is verified as homogeneous.

4. Write any two characteristics of plant kingdom.

Answer:
1. Organisms are multicellular, eukaryotic, and perform autotrophic photosynthesis using chlorophyll.
2. Plant cells are enclosed by a rigid external cellular wall composed of cellulose structural fibers.

5. Write any two roles of honey bee on ecosystem.

Answer:
1. They function as dominant pollinators, facilitating cross-pollination to secure floral and agricultural reproduction.
2. They provide an essential nutrient food resource for multiple insectivorous bird and animal types within the food chain.

6. Mitosis cell division is called equational division. Why?

Answer: Because the total count of chromosomes inside the newly synthesized daughter cells remains identical to the parent cell value.

7. Mention any two benefits gained by farmers on breeding buffalo with Murrah buffalo instead of local species of buffaloes.

Answer:
1. The resulting cross-bred calves exhibit exponentially greater milk production capacities.
2. They show acceleration in weight gain and exhibit ideal attributes for industrial dairy farm environments.

8. Write any two differences between haemophilia and leukemia.

Answer:
1. Haemophilia is a genetic, inherited bleeding disorder caused by missing protein clotting factors, whereas Leukemia is an acquired malignant cancer of bone marrow tissue causing excessive white blood cell production.
2. Haemophilia physically arrests natural blood clotting during external lesions, while Leukemia systematically weakens the primary immune protection of a person.

9. Number of red panda is decreasing in Nepal. What measures can be done to protect the Panda? Write any two measures.

Answer:
1. Securing their fragile natural ecosystems by restricting logging and developing localized bamboo forestry sanctuaries.
2. Instituting harsh structural anti-poaching wildlife legislation and strict legal financial penalties for trafficking.

10. Write any two factors on which gravity on the surface of a heavenly body depends.

Answer:
1. The total concentrated mass of the celestial body (M).
2. The total primary radius from the core to the surface boundary (R).

11. Prove that 'if the earth attracts two bodies placed at the same distance from the centre of the earth with the same force, then their masses are also equal.'

Answer: Let two separate objects have individual masses m1 and m2, located at an equal orbital distance 'd' from Earth's core (mass M).
Applying Newton's gravitational formula provides:
F1 = (G * M * m1) / d^2 and F2 = (G * M * m2) / d^2
Given that these interaction forces are explicitly identical (F1 = F2):
(G * M * m1) / d^2 = (G * M * m2) / d^2
Dividing out the constant factors (G, M, d^2) from both sides isolates:
m1 = m2. This proves their structural mass values are identical.

12. Lamp is lit in the running bicycle by the help of a dynamo but it goes off when bicycle is stopped. Give reason.

Answer: A bicycle dynamo utilizes electromagnetic induction. Moving wheels turn an onboard rotor magnet inside inductive wire loops to transform kinetic rotation into electricity. When the bike stops, magnetic flux stops varying, induced electromotive forces hit zero, and the lamp goes dark.

13. Draw a neat diagram of transformer in mobile charger if input is 220V and output is 3.7V.

Answer: Mobile chargers use a **Step-down transformer** architecture.
- Primary terminal: Wired with a higher relative coil turn volume (Np) hooked to standard 220V AC household current inputs.
- Secondary terminal: Wired with an isolated lower relative turn configuration (Ns) supplying 3.7V AC output lines to charger components.

14. Write a balanced chemical equation between the reactions of a metal of group IIA with a non-metal of group VIIA.

Answer: Balanced chemical combination of Magnesium (Mg, Group IIA) metal reacting with gaseous Chlorine (Cl2, Group VIIA):
Mg + Cl2 -> MgCl2

15. Write the name of members of alkyne which contains an equal number of hydrogen and carbon. Write one uses of it.

Answer: The matching molecule is **Ethyne** (Acetylene gas), with chemical formula **C2H2**.
- Use: Mixed directly with compressed oxygen in commercial oxy-acetylene utility torches to weld or slice iron plates.

16. 'It is dangerous to use chemical pesticides for human health.' Justify this statement by using your two arguments.

Answer:
1. Toxic chemical residues remain attached to crop fields, magnifying across consumption lines to produce neurological toxicity inside human organs.
2. Acute manual exposure triggers sudden systemic metabolic toxicity, severe skin lesions, and permanent hormonal endocrine disruption.

Group 'D' - Long Answer Type Questions (7 x 4 = 28)

17. Write any two positive and negative impacts of digital technology in our daily life.

Answer:
- **Positive:** 1. Facilitates immediate access to specialized data and open-source learning repositories. 2. Automates high-volume database processes and streamlines real-time international text communications.
- **Negative:** 1. Induces inactive routines causing visual impairment, metabolic slowdown, and bad physical posture. 2. Heightens security breaches, data scraping risks, and limits authentic physical human interaction outside networks.

18. Study the given density graph of water and answer:
a) What is the special property of water shown in the figure called?
b) Write the change that takes place in volume of water on heating from 0C to 10C.
c) Write one use of this special property of water.

Answer:
a) This unique physical trait is known as the **Anomalous expansion of water**.
b) When raised from 0C up to 4C, the total volume contracts downward, striking its lowest volume point at exactly 4C. Heating onward from 4C to 10C makes the volume steadily expand.
c) **Use:** It preserves aquatic ecosystems during harsh winters. Floating surface ice provides thermal insulation for the denser 4C water layer remaining fluid underneath, keeping lake animals alive.

19. a) What is an artery? Name the largest artery of human body.
b) Write two causes of high blood pressure.

Answer:
a) An artery is an elastic, thick-walled cardiovascular vessel built to distribute oxygenated blood away from heart cavities. The largest systemic arterial vessel is the **Aorta**.
b) **Causes:** 1. High intake of sodium-heavy, high-cholesterol junk diets along with poor physical movement. 2. Elevated long-term psychological anxieties, active smoking habits, or inherited cardiovascular patterns.

20. Three metal objects A, B, and C of equal mass and volume are heated to the same temperature and placed on a wax vessel. The specific heat capacities are A = 380 J/kgC, B = 470 J/kgC, and C = 910 J/kgC. Answer:
a) Which metal has the highest amount of heat energy?
b) Which metal goes the deepest into the wax vessel?
c) Which metal cools down the fastest?
d) Which metal has the highest amount of heat lose?

Answer:
a) **Metal C** stores the largest net quantity of internal heat energy because it possesses the maximum specific heat capacity rating.
b) **Metal C** travels deepest since its high capacity lets it distribute more total heat to continuously liquefy the wax slab underneath.
c) **Metal A** cools down quickest because its low thermal inertia coefficient allows fast heat radiation loss.
d) **Metal C** experiences the highest absolute scale of total thermal energy loss when normalizing down to room ambient levels.

21. Write two differences between concave and convex lens. The power of the lens used in the spectacles worn by a student is -6.0 D. Calculate the focal length of the lens. Also, mention the type of lens.

Answer:
- **Differences:** 1. Convex lenses are thick centrally and converge light paths, whereas concave variants are thin centrally and diverge light paths. 2. Convex optics form real and inverted projection results, while concave items produce virtual and erect versions.
- **Calculation:** Power P = -6.0 D. Focal length equation: f = 1 / P = 1 / (-6.0) = -0.1667 meters = **-16.67 cm**.
- **Type:** The negative optical value certifies that it is a **Concave lens**.

22. Study the electronic configurations X = 1s2, 2s22p6, 3s23p6, 4s2 and Y = 1s2, 2s22p6, 3s23p5.
Answer:
a) Which one is metal?
b) Write period of the element Y.
c) Write a balanced chemical equation of the compound formed by combination of X and Y.

Answer:
a) **Element X** is an electropositive metal (Calcium, Z=20) because it retains two valence electrons that it easily releases.
b) **Element Y** belongs to the **3rd Period** because its electron cloud fills three principal energy levels.
c) Balanced equation forming a stable binary salt matrix: **X + 2Y -> XY2** (or explicitly written as: **Ca + Cl2 -> CaCl2**)

23. Draw a well-labeled diagram of laboratory preparation of ammonia gas and write down a method to test this gas in the gas jar.

Answer:
- **Lab Synthesis:** An even mixture of dry Ammonium Chloride (NH4Cl) and Calcium Hydroxide (Ca(OH)2) is heated within a downward-tilted boiling tube. The liberated gas is routed through a calcium oxide (CaO) drying cylinder and isolated within an upside-down gas jar via downward air displacement.
- **Testing Routine:** Place a wet red litmus paper strip at the opening of the collecting jar. The highly alkaline nature of expanding ammonia turns the indicator strip from **red to blue** instantly.